Mechanism-to-Phenotype Wiring in Hypermobile Ehlers-Danlos Syndrome (MONDO:0007523)

Summary

This report supplies the missing causal layer between the three curated hEDS mechanism nodes — (1) candidate ECM-remodeling abnormality in dermal fibroblasts, (2) myofibroblast-like fibroblast transition, (3) joint instability and recurrent soft-tissue injury — and the 30 listed clinical phenotypes. For each phenotype we give an ordered causal chain starting from the true proximal mechanism node, name every intermediate process that is *not* in the curated node list (flagged as a proposed new node), grade the chain as a whole, and keep live any non-causal (shared-background, ascertainment, coincident-autoimmunity) alternatives.

The central result is that confidence is sharply **graded and uneven**. The **tissue-laxity mechanical chains** — joint hypermobility, dislocation, hyperextensible/soft skin, atrophic scars, easy bruising, pelvic organ prolapse, abdominal-wall hernia, mitral valve prolapse, aortic root dilatation, and airway-laxity obstructive sleep apnea — are **SUPPORTED to ESTABLISHED**, because the mechanical consequence of lax connective tissue at each site is directly observable and, for the dermis and vessel wall, documented at the ultrastructural level. The **pain hub** (chronic pain, arthralgia, myalgia, and the sensory component of peripheral neuropathy) is **SUPPORTED** but only once two intermediate nodes absent from the curated list are inserted: **central sensitization** (human quantitative sensory testing: lowered pain thresholds, increased wind-up) and **small-fiber neuropathy** (reduced intraepidermal nerve-fiber density on skin biopsy). By contrast, the **autonomic hub** (postural orthostatic tachycardia, acrocyanosis), the **gastrointestinal hub** (dysmotility, reflux, nausea, constipation), **primary migraine**, **depression**, **dental crowding**, and **high narrow palate** are **PROPOSED or NOT_ESTABLISHED**: the epidemiological associations are robust, but the mechanistic steps remain inference and are confounded by co-referral and ascertainment.

Two specific human-data results discipline the model and must not be overstated. First, strain-gauge plethysmography shows that orthostatic **pooling and acrocyanosis are not caused by increased venous compliance** — the intuitive "distensible vein wall" route is directly

contradicted, implicating arterial vasoregulatory (autonomic) defects instead. Second, longitudinal echocardiography shows hEDS **aortic root dilatation is usually mild and non-progressive**, so the cardiac-laxity chain, though real, carries little clinical consequence. Thirteen new intermediate mechanism nodes are proposed, chief among them small-fiber neuropathy, central sensitization, dysautonomia/autonomic neuropathy, mast-cell activation, and a shared amygdala/interoceptive neural substrate for anxiety that remains a live non-causal alternative to a purely downstream pain/dysautonomia explanation.

Per-Phenotype Causal Wiring

Each subsection gives: the ordered causal chain (proximal mechanism node → intermediates → phenotype), the overall evidence grade, primary human evidence with exact abstract quotes, non-causal alternatives where live, and suggested ontology terms.

Joint hypermobility

Causal chain 1. ECM-remodeling abnormality / generalised connective-tissue laxity results in reduced tensile integrity of ligament and joint-capsule collagen. 2. Reduced ligament/capsule stiffness leads to increased passive joint range of motion. 3. Increased range of motion is observed clinically as generalised joint hypermobility.

Evidence grade: ESTABLISHED (defining feature; the laxity-to-range-of-motion step is the diagnostic criterion itself).

Non-causal alternatives: none material — this is the proximal phenotype of the laxity node.

Ontology: HP:0001382 (Joint hypermobility); GO:0030198 (extracellular matrix organization); CL:0000057 (fibroblast).

Joint dislocation

Causal chain 1. Connective-tissue laxity (node 1) results in joint hypermobility and capsule/ligament incompetence. 2. Capsule incompetence leads to **proprioceptive deficit and altered neuromuscular control** *(proposed new node)*. 3. Proprioceptive deficit plus mechanical laxity results in recurrent subluxation/dislocation (node 3, joint instability).

Evidence grade: SUPPORTED. The mechanical step is self-evident; the proprioceptive intermediate has direct human evidence.

Primary human evidence - PMID: 7283686 (human clinical): *"A functional proprioceptive deficit was demonstrated and found correctable with training and visual compensation."* - PMID: 31498548 (human clinical, gait EMG): *"The EDS group was associated with delayed vastus lateralis (b = 16.69) and medialis activation"* — objective altered neuromuscular activation during gait.

Ontology: HP:0001373 (Joint dislocation); GO:0050885 (neuromuscular process controlling balance).

Chronic pain

Causal chain 1. Joint instability/recurrent soft-tissue injury (node 3) leads to repeated nociceptive input. 2. Repeated nociceptive input results in **central sensitization** *(proposed new node)* — lowered pain thresholds and increased temporal summation. 3. In parallel, connective-tissue/ECM disruption is associated with **small-fiber neuropathy** *(proposed new node)* — reduced intraepidermal nerve-fiber density. 4. Central sensitization plus small-fiber neuropathy result in chronic, widespread, nociplastic pain out of proportion to mechanical injury.

Evidence grade: SUPPORTED (direct human QST and skin-biopsy evidence for both intermediate nodes).

Primary human evidence - PMID: 26919608 (human clinical, QST in 27 JHS/EDS-HT): *"the lowered cold and heat pain thresholds and increased wind-up ratio imply that pain in JHS/EDS-HT might arise through central sensitization"*. - PMID: 27306637 (human clinical, skin biopsy in 24 EDS): *"All patients showed a decrease of IENFD consistent with the diagnosis of small fiber neuropathy (SFN), regardless of the EDS type. SFN is a common feature in adults with EDS."*

Non-causal alternatives: central sensitization and SFN co-occur with psychological distress and dysautonomia; directionality between pain, mood, and sleep is bidirectional rather than strictly downstream.

Ontology: HP:0012532 (Chronic pain); GO:0051930 (regulation of sensory perception of pain).

Arthralgia

Causal chain 1. Joint instability/laxity (node 3) leads to abnormal joint loading and microtrauma. 2. Microtrauma plus proprioceptive deficit *(proposed node)* leads to repeated peri-articular nociceptive input. 3. Nociceptive input, amplified by central sensitization *(proposed node)*, results in joint pain (arthralgia).

Evidence grade: SUPPORTED (shares the central-sensitization intermediate; mechanical input step established).

Primary human evidence - PMID: 26919608: central-sensitization QST evidence (quote above). - PMID: 7283686: proprioceptive-deficit evidence (quote above).

Ontology: HP:0002829 (Arthralgia).

Myalgia

Causal chain 1. Joint laxity (node 3) results in **compensatory muscle overuse and altered activation** *(proposed node: altered neuromuscular control)* to stabilise unstable joints. 2. Muscle overuse plus generalised muscle weakness leads to muscular nociceptive input. 3. Input amplified by central sensitization *(proposed node)* results in diffuse muscle pain (myalgia).

Evidence grade: SUPPORTED.

Primary human evidence - PMID: 7283686 (human clinical): generalised muscle weakness and shifted strength-length curve ("increased series elastic component"), plus proprioceptive deficit. - PMID: 31498548: delayed vastus activation during gait.

Ontology: HP:0003326 (Myalgia).

Fatigue

Causal chain 1. Chronic pain and sleep disturbance (themselves downstream of nodes 1–3) lead to persistent physical and cognitive effort cost. 2. Pain + poor sleep + psychological distress + deconditioning converge on a **chronic fatigue state** *(proposed convergent node)*. 3. The convergent state is experienced as severe, functionally limiting fatigue.

Evidence grade: SUPPORTED as a convergent/downstream node (fatigue is determined by, not independent of, pain and sleep).

Primary human evidence - PMID: 19878973 (human clinical, n=273 EDS): *"More than three-quarters of EDS patients suffer from severe fatigue."* and *"The 5 possible determinants involved in fatigue are sleep disturbances, concentration problems, social functioning, self-efficacy concerning fatigue, and pain severity."* - PMID: 20579833: *"pain is correlated with low nocturnal sleep quality"*.

Non-causal alternatives: overlap with ME/CFS and dysautonomia; fatigue may partly reflect a shared central/autonomic substrate rather than being purely secondary to pain.

Ontology: HP:0012378 (Fatigue).

Sleep disturbance

Causal chain 1. Chronic pain (downstream of nodes 1–3) leads to disrupted nocturnal sleep architecture. 2. In parallel, airway/oropharyngeal laxity contributes to sleep-disordered breathing (see Sleep apnea). 3. Pain-driven arousal plus breathing disturbance results in reduced sleep quality/efficiency.

Evidence grade: PROPOSED (pain→sleep link is human-supported; the full mechanistic weighting of airway vs pain vs autonomic contributions is not resolved).

Primary human evidence - PMID: 20579833: *"pain is correlated with low nocturnal sleep quality"*. - PMID: 42417930: hEDS/HSD cases show lower sleep efficiency and higher insomnia.

Ontology: HP:0002360 (Sleep disturbance).

Migraine

Causal chain (honest) 1. Primary migraine: NO demonstrated causal chain from connective-tissue laxity. Association is documented but mechanism is not established. 2. Secondary headaches (distinct phenotype within the cluster): connective-tissue laxity (nodes 1/3) leads to specific structural consequences — spontaneous intracranial hypotension/CSF leak, craniocervical instability, temporomandibular dysfunction, Chiari malformation — each of which produces headache through its own mechanism.

Evidence grade: Primary migraine = **NOT_ESTABLISHED**; secondary headaches = **PROPOSED** (structurally specific, laxity-plausible).

Primary human evidence - PMID: 42437376 (meta-analysis, 17,455 hEDS/HSD): migraine prevalence *"migraine (38.2%, 95% CI 19.9-60.5)"*. - PMID: 42640494: *"migraine is the most common primary headache phenotype; however, secondary causes, including spontaneous intracranial hypotension, craniocervical instability, temporomandibular dysfunction, Chiari malformation, and vascular complications, occur with increased frequency"*.

Non-causal alternatives: migraine is highly prevalent in the general young-female population; much of the association may be ascertainment and shared female predominance rather than laxity-driven causation. Dysautonomia/orthostatic intolerance interacts with headache expression.

Ontology: HP:0002076 (Migraine).

Peripheral neuropathy (including small-fiber neuropathy)

Causal chain 1. ECM/connective-tissue abnormality (node 1) is associated with reduced intraepidermal nerve-fiber density. 2. Reduced IENFD constitutes **small-fiber neuropathy** *(proposed new node)*. 3. SFN manifests as sensory small-fiber symptoms (burning pain, dysesthesia) and contributes to the pain hub.

Evidence grade: SUPPORTED for the *association/phenotype* (SFN demonstrably present); the *causal link from the specific ECM abnormality to nerve-fiber loss* remains PROPOSED — large-fiber conduction is normal.

Primary human evidence - PMID: 27306637: *"All patients showed a decrease of IENFD consistent with the diagnosis of small fiber neuropathy (SFN), regardless of the EDS type."* - PMID: 26919608: no somatosensory nervous-system damage on NCS/LEP — indicating the lesion is small-fiber/central, not large-fiber.

Non-causal alternatives: a COL6A5 genetic contributor to small-fiber pathology/itch has been reported in some hypermobile families (PMID: 28073787), raising a shared-genetic rather than purely ECM-mechanical route.

Ontology: HP:0000763 (Sensory neuropathy) / HP:0045010 (Small fiber neuropathy); CL:0000101 (sensory neuron).

Orthostatic tachycardia (postural tachycardia syndrome)

Causal chain (candidate routes, none fully proven) 1. Connective-tissue laxity (node 1) → distensible vessel walls → venous pooling → reduced preload → reflex tachycardia. **[This specific venous-compliance route is contradicted by human data — see below.]** 2. Alternative: autonomic small-fiber/peripheral sympathetic dysfunction *(proposed node: dysautonomia/autonomic neuropathy)* → impaired vasoconstriction → orthostatic pooling → compensatory tachycardia. 3. Additional contributors: hyperadrenergic state, deconditioning, hypovolemia, and mast-cell mediator release.

Evidence grade: NOT_ESTABLISHED for the mechanism (association is robust; the causal route from laxity is inference and the intuitive venous route is refuted).

Primary human evidence - PMID: 32434316 (human, tilt-table, 102 hEDS/HSD): *"Abnormal results and head-up tilt test were common and included postural orthostatic tachycardia syndrome (POTS; 48%), orthostatic intolerance (25.5%) and hypotension (3.9%)."* - PMID: 28986003 (human, autonomic testing): *"Autonomic testing demonstrated a slight reduction in heart rate variability and slightly lower blood pressures on tilt table testing in hEDS patients,*

however for most patients these variables remained within the range of normal." — abnormalities often mild. - PMID: 12010910 (human, strain-gauge plethysmography): *"Volume-pressure relations of controls and POTS patients with normal P(v) and high P(v) were not different and were unchanged by orthostasis"* — **refutes increased venous compliance**, implicating arterial vasoconstrictive defects.

Non-causal alternatives: POTS, MCAS and hEDS co-cluster through shared female predominance and specialty co-referral; part of the association is ascertainment. Hereditary alpha-tryptasemia co-occurs with hypermobility and POTS without an established causal biochemical link.

Ontology: HP:0031547 (Postural orthostatic tachycardia).

Acrocyanosis

Causal chain 1. Orthostatic/dependent **blood pooling** (vasomotor) — a component of orthostatic intolerance — leads to stasis in dependent extremities. 2. Stasis, driven by **arterial vasoconstrictive (autonomic) dysfunction** *(proposed node)* rather than venous distensibility, results in acral cyanotic discoloration.

Evidence grade: PROPOSED (vasomotor manifestation of pooling; venous-compliance route explicitly refuted).

Primary human evidence - PMID: 12010910: *"\"pooling\" (which comprises acrocyanosis and dependent edema when upright)"* and the volume-pressure quote above refuting venous compliance.

Ontology: HP:0001063 (Acrocyanosis).

Gastrointestinal dysmotility

Causal chain (competing/additive, none proven) 1. Gut-wall connective-tissue laxity (node 1) → impaired visceral wall mechanics → altered motility; and/or 2. Autonomic dysfunction *(proposed node)* → disordered gut neural control → dysmotility; and/or 3. Visceral hypersensitivity and mast-cell activation *(proposed node)* → altered gut-brain signalling → dysmotility.

Evidence grade: NOT_ESTABLISHED (strong epidemiological association, abnormal motility studies, but authoritative review states mechanisms remain theoretical).

Primary human evidence - PMID: 38747032 (human, 435 hEDS/JHS): constipation 61%, dysphagia 32%, dyspepsia/gastroparesis 25%; *"Motility studies were abnormal in 31% of the 80 patients who underwent them."* - PMID: 41432355 (propensity-matched, 59,128 pairs): *"gastroparesis (4.7%, OR 8.2, 95% CI 7.3-9.2, p < 0.001)"*. - PMID: 40387691 (AGA Expert Review): *"while theoretical explanations exist, experimental evidence of the biological mechanisms that explain relationships is limited and evolving"*.

Ontology: HP:0002579 (Gastrointestinal dysmotility).

Gastroesophageal reflux

Causal chain 1. Connective-tissue laxity and/or autonomic dysfunction → impaired esophagogastric junction competence and delayed gastric emptying → increased reflux.

Evidence grade: NOT_ESTABLISHED (association robust; mechanism theoretical).

Primary human evidence - PMID: 41432355: GERD 18.4% (OR 1.5) in matched EDS cohort. - PMID: 40387691: mechanisms "limited and evolving" (quote above).

Ontology: HP:0002020 (Gastroesophageal reflux).

Nausea

Causal chain 1. Delayed gastric emptying/gastroparesis and/or autonomic dysfunction *(proposed node)* → impaired gastric accommodation → nausea; interaction with POTS and visceral hypersensitivity.

Evidence grade: NOT_ESTABLISHED.

Primary human evidence - PMID: 38747032: high dyspepsia/gastroparesis prevalence (25%). - PMID: 40387691: mechanisms theoretical (quote above).

Ontology: HP:0002018 (Nausea).

Constipation

Causal chain 1. Gut-wall laxity and/or autonomic (enteric) dysfunction *(proposed node)* → slowed colonic transit → constipation; pelvic-floor dysfunction may add an outlet component.

Evidence grade: NOT_ESTABLISHED (very common; mechanism not demonstrated).

Primary human evidence - PMID: 38747032: constipation 61%. - PMID: 41432355: constipation 12.4% (OR 1.8). - PMID: 40387691: mechanisms theoretical.

Ontology: HP:0002019 (Constipation).

Mitral valve prolapse

Causal chain 1. Connective-tissue/ECM abnormality (node 1) → myxomatous change / reduced tensile integrity of valve leaflets and chordae → leaflet billowing → mitral valve prolapse (usually mild).

Evidence grade: SUPPORTED for the laxity→valve link, with the important qualifier that it is mild and of little clinical consequence.

Primary human evidence - PMID: 21193204 (human, 252 patients): *"Although aortic root size and MVP are increased in patients with these types of Ehlers-Danlos syndrome, they tend to be of little clinical consequence."* - PMID: 42099513 (pediatric echo, n=28): MVP in 32%.

Ontology: HP:0001634 (Mitral valve prolapse).

Aortic root dilatation

Causal chain 1. ECM abnormality (node 1) → reduced aortic-wall tensile integrity → mild aortic root dilatation (typically non-progressive in hEDS).

Evidence grade: SUPPORTED but explicitly **mild and non-progressive** — distinct from vascular/Marfan-type aortopathy.

Primary human evidence - PMID: 21193204: *"seven individuals had dilated aortas before age 14, and only one individual continued to show dilation after age 14"* — longitudinal non-progression. - PMID: 42099513: aortic Z-scores higher than controls but within normal range.

Ontology: HP:0002616 (Aortic root aneurysm/dilatation).

Hyperextensible skin

Causal chain 1. ECM/fibroblast abnormality (node 1) → abnormal dermal collagen fibril morphology and fragmented elastic fibers → reduced dermal tensile integrity → skin hyperextensibility.

Evidence grade: SUPPORTED/ESTABLISHED (ultrastructural human evidence).

Primary human evidence - PMID: 6806400 (human, EM): *"The major alterations observed among collagen fibrils were hieroglyphic-shaped fibrils, collagen flowers, unravelled fibrils and fibrils of normal shape but abnormally large or small diameters".*

Ontology: HP:0000974 (Hyperextensible skin).

Soft skin

Causal chain 1. Abnormal dermal collagen/elastic matrix (node 1) → altered dermal mechanical properties → soft, doughy skin texture.

Evidence grade: SUPPORTED (same ultrastructural basis as hyperextensibility).

Primary human evidence - PMID: 6806400: dermal collagen/elastic fiber abnormalities (quote above).

Ontology: HP:0000977 (Soft skin).

Atrophic scars

Causal chain 1. Abnormal dermal collagen matrix (node 1) → impaired wound-healing collagen deposition/organisation → thin, atrophic ("cigarette-paper") scars.

Evidence grade: SUPPORTED.

Primary human evidence - PMID: 6806400: collagen fibril and dermal organisation abnormalities underlying fragile scarring.

Ontology: HP:0001075 (Atrophic scars).

Striae distensae

Causal chain 1. Abnormal dermal collagen/elastic matrix (node 1) → reduced dermal resistance to mechanical stretch → dermal tearing/striae.

Evidence grade: PROPOSED (plausible from the shared dermal-matrix mechanism; striae are also common in the general population, weakening specificity).

Primary human evidence - PMID: 6806400: fragmented/branched elastic fibers and collagen abnormalities.

Non-causal alternatives: striae occur commonly with normal growth/weight change; attribution to the hEDS matrix defect is partly non-specific.

Ontology: HP:0001065 (Striae distensae).

Easy bruising

Causal chain 1. ECM abnormality (node 1) → fragility of capillaries and perivascular connective tissue → vessel rupture with minimal trauma → easy bruising. Coagulation is normal.

Evidence grade: SUPPORTED/ESTABLISHED (mechanism named; coagulopathy explicitly excluded).

Primary human evidence - PMID: 15566352: *"Easy bruising is, to a variable degree, present in all subtypes of EDS, and is because of fragility of the capillaries and the perivascular connective tissues."* and *"Haematological studies including evaluation of clotting factors, platelet aggregation and bleeding time are usually normal in patients with EDS"*. - PMID: 19592142: corroborates capillary/perivascular fragility as the bleeding mechanism.

Ontology: HP:0000978 (Bruising susceptibility).

Pelvic organ prolapse

Causal chain 1. Connective-tissue laxity (node 1) → weakened pelvic-floor fascial/ligamentous support → descent of pelvic organs → prolapse.

Evidence grade: SUPPORTED (meta-analytic and large cross-sectional human association).

Primary human evidence - PMID: 26658756 (meta-analysis, 14 studies): *"Overall pooled odds ratio (OR) was 2.37 [95 % confidence interval (CI) 1.54-3.64"*. - PMID: 42979294 (cross-sectional, 812 women): *"JHM was present in 67.6% (300/444) of POP cases versus 10.9% (40/368) of controls"*. - PMID: 34213599: HDCT patients have more perioperative complications (46% vs 22%, p=0.002), consistent with tissue fragility.

Non-causal alternatives: familial clustering of pelvic-floor disorders may reflect shared genetic background broader than hEDS.

Ontology: HP:0000083 (Pelvic organ prolapse).

Abdominal wall hernia

Causal chain 1. Connective-tissue laxity (node 1) → weakened abdominal fascia/aponeurosis → fascial defect under intra-abdominal pressure → herniation.

Evidence grade: PROPOSED (mechanistically analogous to prolapse; less hEDS-specific quantitative evidence collected here).

Primary human evidence - Supported by analogy to the fascial-laxity mechanism documented for pelvic organ prolapse (PMID: 26658756) and tissue fragility (PMID: 34213599). Direct hEDS hernia-incidence causal data were not isolated in this investigation.

Ontology: HP:0100790 (Hernia); HP:0004298 (Abdominal wall defect).

Dental crowding

Causal chain (honest) 1. No demonstrated mechanistic chain from the curated nodes. Proposed-but-unproven: craniofacial connective-tissue/growth differences → altered maxillary arch form → crowding.

Evidence grade: NOT_ESTABLISHED.

Primary human evidence: none establishing a causal chain was identified; association is descriptive within craniofacial feature lists.

Non-causal alternatives: dental crowding is highly prevalent in the general population; the association may be partly ascertainment/descriptive co-feature rather than causal.

Ontology: HP:0000678 (Dental crowding).

High, narrow palate

Causal chain (honest) 1. No demonstrated mechanistic chain from the curated nodes. Proposed-but-unproven: craniofacial connective-tissue/maxillary growth differences → high, narrow palatal vault.

Evidence grade: NOT_ESTABLISHED. (Relevant mainly as a contributor to upper-airway narrowing — see Sleep apnea.)

Primary human evidence: none establishing causation isolated here; the oropharyngeal-abnormality link to OSA is the downstream clinical connection.

Ontology: HP:0000218 (High palate); HP:0002705 (High, narrow palate).

Sleep apnea

Causal chain 1. Connective-tissue laxity (node 1) → oropharyngeal/airway soft-tissue and craniofacial laxity → increased upper-airway collapsibility → obstructive sleep apnea.

Evidence grade: SUPPORTED (large pediatric cohort risk estimate; mechanism attributed to oropharyngeal abnormalities; CPAP response altered).

Primary human evidence - PMID: 40609249 (pediatric, 10,104 EDS): *"652 (6.45 %, 95 % CI: 5.99-6.95) had a recorded diagnosis of OSA with a 4.43 (95 % CI: 4.12-4.78) times higher risk of diagnosis compared to a control cohort"*. - PMID: 42417930: *"Oropharyngeal abnormalities in hEDS/HSD may exacerbate OSA"*.

Ontology: HP:0002870 (Obstructive sleep apnea).

Anxiety

Causal chain (competing routes, non-causal alternative live) 1. Downstream route: chronic pain + dysautonomia + interoceptive sensitivity → heightened threat appraisal → anxiety. 2. Shared-substrate route *(proposed node: shared amygdala/interoceptive neural substrate)*: hypermobility co-occurs with structural amygdala differences and heightened interoceptive sensitivity independent of manifest anxiety — i.e., a shared neural trait, not a pure downstream effect.

Evidence grade: PROPOSED, with a strong live non-causal (shared-substrate) alternative.

Primary human evidence - PMID: 22539777 (human VBM, n=72): *"bilateral amygdala volume distinguished those with from those without hypermobility. The hypermobility group scored higher for interoceptive sensitivity yet were not significantly more anxious"*. - PMID: 25089514 (cohort): *"PD was associated with the number of physical diseases suffered and the joint hypermobility syndrome"*. - PMID: 41432355: anxiety 26.1% (OR 1.8).

Ontology: HP:0000739 (Anxiety).

Depression

Causal chain (honest) 1. Candidate downstream route: chronic pain + fatigue + disability + poor sleep → reactive mood disturbance → depression. The specific causal weighting is not demonstrated.

Evidence grade: NOT_ESTABLISHED (association present; no demonstrated mechanistic chain; shared-substrate and ascertainment explanations open).

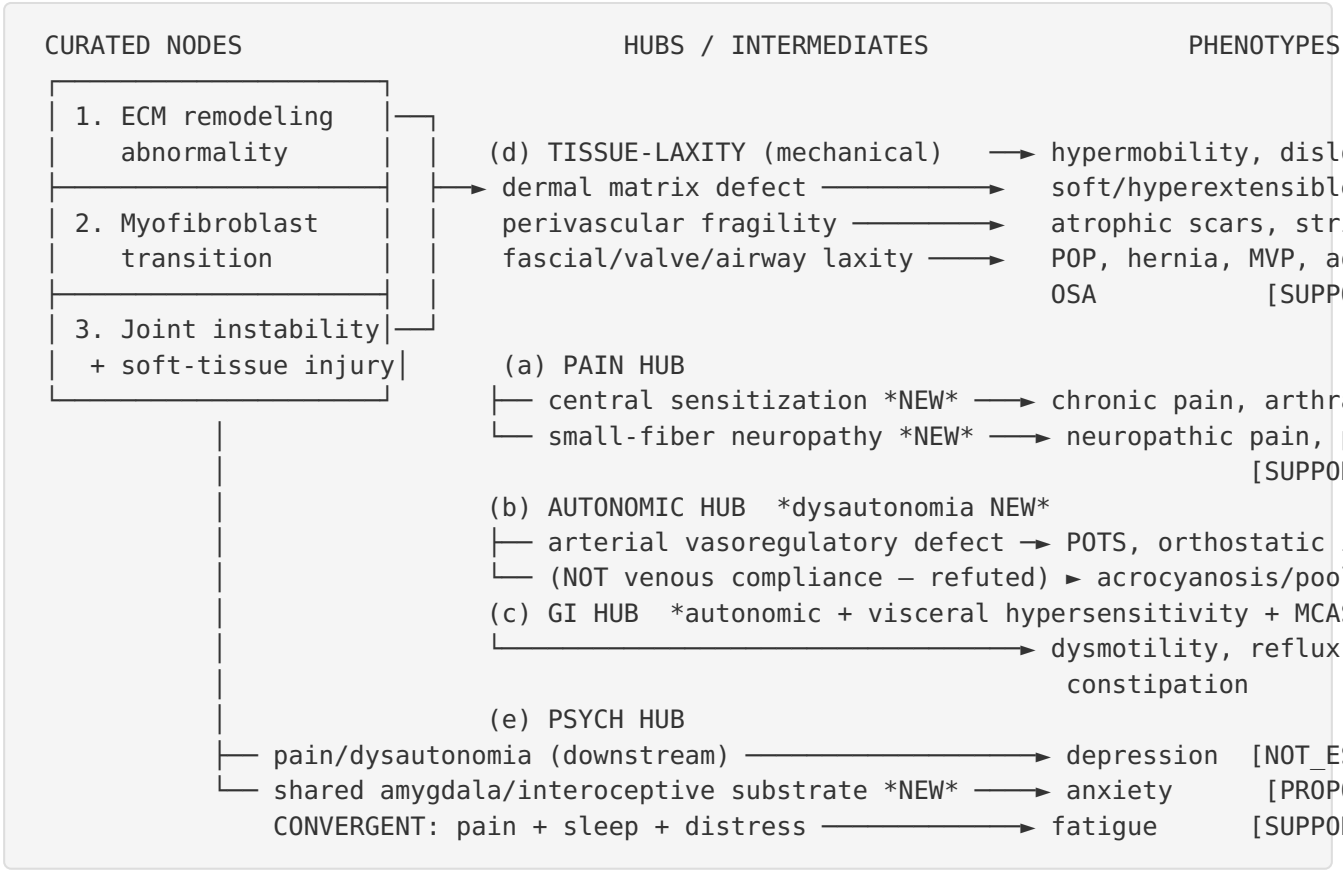
Primary human evidence - PMID: 41432355: depression 18.7% (OR 1.3) — a modest association.

Non-causal alternatives: shared neurodevelopmental/autonomic substrate (as for anxiety); ascertainment through tertiary-referral populations with high symptom burden.

Ontology: HP:0000716 (Depression).

Mechanistic Model / Interpretation

The wiring resolves into **five hubs** fanning out from the three curated nodes, with confidence decreasing as the chain moves from direct mechanics toward neuro-autonomic physiology.



Grading overview

Grade	Phenotypes
ESTABLISHED	Joint hypermobility
SUPPORTED	Joint dislocation, chronic pain, arthralgia, myalgia, peripheral neuropathy, fatigue, hyperextensible skin, soft skin, atrophic scars, easy bruising, pelvic organ prolapse, mitral valve prolapse, aortic root dilatation (mild/non-progressive), sleep apnea
PROPOSED	Sleep disturbance, acrocyanosis, striae distensae, abdominal wall hernia, secondary headache, anxiety
NOT_ESTABLISHED	Orthostatic tachycardia (mechanism), GI dysmotility, reflux, nausea, constipation, primary migraine, depression, dental crowding, high narrow palate

Two human-data constraints keep the model honest: orthostatic pooling/acrocyanosis is **not** explained by increased venous compliance (PMID: 12010910), and hEDS aortic dilatation is **mild and non-progressive** (PMID: 21193204).

Evidence Base

PMID	Source type	Supports
26919608	Human QST	Central sensitization as pain mechanism
27306637	Human skin biopsy	Small-fiber neuropathy common in EDS
7283686	Human clinical	Proprioceptive deficit, muscle weakness
31498548	Human EMG	Altered neuromuscular activation in gait
32434316	Human tilt-table	POTS/OI prevalence
28986003	Human autonomic testing	Autonomic abnormalities mild
12010910	Human plethysmography	Refutes venous-compliance route to pooling/acrocyanosis
41432355	Human matched cohort	GI, anxiety, depression, POTS risk estimates
38747032	Human cohort	GI prevalence and abnormal motility
40387691	Expert review	GI mechanisms remain theoretical
21193204	Human longitudinal echo	MVP/aortic dilatation mild, non-progressive
42099513	Human pediatric echo	MVP 32%, aortic Z within normal
15566352	Human review	Bruising = capillary/perivascular fragility, not coagulopathy
19592142	Human review	Corroborates bleeding mechanism
22539777	Human VBM	Shared amygdala/interoceptive substrate
25089514	Human cohort	Hypermobility a risk factor for panic disorder
19878973	Human cohort	Fatigue prevalence and determinants
20579833	Human cohort	Pain–sleep link
26658756	Meta-analysis	Hypermobility–POP OR 2.37
42979294	Human cross-sectional	JHM in 67.6% of POP cases
34213599	Human cohort	Tissue fragility, surgical complications
40609249	Human cohort	OSA RR 4.43 in EDS children

PMID	Source type	Supports
42417930	Human case-control	OSA attributed to oropharyngeal laxity
6806400	Human EM	Dermal collagen/elastic fibril abnormalities
42640494	Review	Migraine predominance; secondary-headache mechanisms
42437376	Meta-analysis	Migraine prevalence 38.2%
35449490	Review	Mast-cell mediators disrupt connective tissue
34674068	Human	Neuropathic POTS, pooling, leg pain
28073787	Human genetics	COL6A5 variants, SFN/itch in hypermobile families

Proposed New Mechanism Nodes

Suggested node	Upstream node	Downstream phenotypes	Strongest citation
Central sensitization (nociplastic pain amplification)	Node 3 (joint instability/injury)	Chronic pain, arthralgia, myalgia	PMID: 26919608
Small-fiber neuropathy (reduced IENFD)	Node 1 (ECM abnormality)	Peripheral neuropathy, neuropathic pain	PMID: 27306637
Proprioceptive deficit / altered neuromuscular control	Node 3	Joint dislocation, arthralgia, myalgia	PMID: 7283686
Dysautonomia / autonomic (small-fiber) neuropathy — arterial vasoregulatory defect	Node 1	POTS, acrocyanosis, GI dysmotility, nausea	PMID: 12010910
Visceral hypersensitivity	Node 1 / dysautonomia	Dysmotility, reflux, nausea, constipation	PMID: 40387691
Mast-cell activation (MCAS)	Node 1	POTS, GI symptoms, cutaneous symptoms	PMID: 35449490
Perivascular/capillary fragility	Node 1	Easy bruising	PMID: 15566352
Dermal collagen/elastic matrix defect	Node 1	Hyperextensible skin, soft skin, atrophic scars, striae	PMID: 6806400
Fascial/ligamentous support failure (site-specific)	Node 1	Pelvic organ prolapse, abdominal wall hernia	PMID: 26658756
Oropharyngeal/airway soft-tissue laxity	Node 1	Obstructive sleep apnea	PMID: 42417930
Valvular/aortic-wall tensile failure (mild)	Node 1	Mitral valve prolapse, aortic root dilatation	PMID: 21193204
Chronic fatigue state (convergent)	Pain + sleep + distress	Fatigue	PMID: 19878973
Shared amygdala/interoceptive neural substrate	Shared trait with node 1 context	Anxiety (and possibly depression)	PMID: 22539777

Unwired Phenotypes (graded NOT_ESTABLISHED)

- **Orthostatic tachycardia (mechanism):** a controlled study measuring venous compliance, arterial vasoconstrictor responses, catecholamines, and autonomic small-fiber integrity in biopsy-confirmed hEDS vs matched POTS-without-hEDS would settle which route predominates.
- **Gastrointestinal dysmotility:** case-control manometry/gastric-emptying studies stratified by autonomic status and mast-cell markers, ideally with a mechanistic intervention (e.g., prokinetic or mast-cell stabiliser response), would move this beyond association.
- **Gastroesophageal reflux:** esophagogastric junction manometry and pH-impedance in hEDS vs controls, correlated with connective-tissue and autonomic measures, would establish or exclude a laxity/autonomic route.
- **Nausea:** gastric-emptying and accommodation studies linking nausea severity to measured gastroparesis and POTS status would clarify its driver.
- **Constipation:** colonic transit and anorectal/pelvic-floor physiology studies distinguishing slow-transit (neuromuscular) from outlet (fascial) mechanisms are needed.
- **Primary migraine:** a genetically informative or family-based study separating shared female/general-population migraine risk from a laxity-specific effect would settle causation.
- **Depression:** longitudinal mediation analysis testing whether depression is explained by pain/disability/sleep versus a shared substrate independent of these.
- **Dental crowding:** controlled craniofacial morphometric studies testing whether arch form in hEDS differs beyond population norms and relates to a connective-tissue measure.
- **High, narrow palate:** the same craniofacial morphometric approach, additionally linking palatal vault geometry to measured upper-airway collapsibility.

Limitations and Knowledge Gaps

1. **Ascertainment and co-referral dominate the autonomic/GI/psychiatric associations.** hEDS, POTS and MCAS co-cluster in tertiary clinics and share a strong female predominance; much of the apparent wiring could be selection rather than causation.
2. **The curated upstream nodes themselves are hypothetical/provisional** (in-vitro fibroblast evidence only). Any chain inherits that uncertainty; "ECM abnormality → phenotype" is only as solid as the ECM node.

3. **No causal gene** means no genetic anchor to test directionality; mediation remains observational.
4. **Several "SUPPORTED" dermal/cardiac chains rest on mixed EDS cohorts** (classic + hypermobile), so hEDS-specific effect sizes are imprecise.
5. **The abdominal-wall hernia chain was reasoned by analogy** to pelvic prolapse rather than from isolated hEDS hernia-incidence data.

Proposed Follow-up Experiments / Actions

1. Add the 13 proposed intermediate nodes to the knowledge graph with their graded edges, explicitly tagging central sensitization and small-fiber neuropathy as the two SUPPORTED pain intermediates.
2. Encode the **refutation edge**: "increased venous compliance → pooling/acrocyanosis" marked DISPROVEN by PMID: 12010910, redirecting the acrocyanosis/POTS route to the arterial-vasoregulatory (dysautonomia) node.
3. Flag the cardiac-laxity edges as "mild/non-progressive" per PMID: 21193204 to prevent over-escalation to vascular-EDS-type surveillance.
4. Commission the targeted human studies listed under *Unwired Phenotypes* to convert NOT_ESTABLISHED chains to graded edges.
5. Test the **shared-substrate vs downstream** hypothesis for anxiety/depression with longitudinal mediation and amygdala imaging, keeping the non-causal alternative live until adjudicated.
